EE2004 lab session, week7, section Thursday

main.c

```
1  unsigned int a;
2  unsigned char b1,b2,b3,b4;
3
4  int *mem_loc;//pointer to the memory location
5
6  int main(void)
7  {
8  mem_loc= (int *) 0x20000040; //memory location to be reached
9
10 *mem_loc=0x87654321; //modify the content of memory location
11
12 a=*mem_loc;//acquire the content of memory location
13
14 //a>>n operation right shifts the content of variable a by n bits.
15
16 b1=a;
17 b2=a>>8;
18 b3=a>>16;
19 b4=a>>24;
20
21 mem_loc+=1;//point to the next location
22
23    for(;;);
24 }
```

Expressions | Modules | Live Expressions

Expression	Type	Value
a	unsigned i...	0x87654321 (Hex)
mem_loc	int *	0x20000044
b1	unsigned ...	0x21 (Hex)
b2	unsigned ...	0x43 (Hex)
b3	unsigned ...	0x65 (Hex)
b4	unsigned ...	0x87 (Hex)
Add new ex		

The above code is an example on how to reach a specific memory location in C language using pointers. The only constraint here is that we can reach a memory location having content of 32 bits. To reach 8-bit values within this 32-bit content, we should apply the right shift operation.

Task:

a) Sixteen characters are saved in memory starting from 0x20000040.
b) The content will be as follows.

0xF0, 0xF1, 0xF2, 0xF3, 0xF4, 0xF5, 0xF6, 0xF7, 0xF8, 0xF9, 0xFA, 0xFB, 0xFC, 0xFD, 0xFE, 0xFF

c) Find the average value of the odd and even numbers by reaching the memory content.

Hint:

- a=a<<8; left shifts the bits in variable a by 8.
- b1 & 0x01 operation is done to apply bitwise AND operation between unsigned char variable b1 and hexadecimal number 0x01
- a|=b1 operation is done to apply bitwise OR operation between unsigned char variable a and b1. The result is saved in variable a.